

Research article

Suppressing thin-wall instability through self-pressurizing of the medium: A novel gas-liquid low-pressure tube forming technology

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ABSTRACT

A novel gas-liquid hybrid low-pressure forming (GLLF) process is proposed, which employs a compressible gas-liquid medium with a tunable compression ratio to eliminate pressure fluctuations and produce an optimal pressure profile for plastic deformation, thereby effectively suppressing compression instability. We develop a gas-liquid hybrid medium compression model based on Soave-Redlich-Kwong (SRK) real-gas equation and solubility theory, quantifying the roles of initial gas fraction, initial pressure, and volume compression ratio on pressure evolution. Experimental validation yields a maximum prediction error of only 2.17 %. A critical support pressure curve for thin-walled tubes at different compression ratios was experimentally measured to characterize wrinkling instability. A strategy for effectively suppressing wrinkling by adjusting the gas-liquid ratio and initial pressure was proposed. Experiments on 6061-O aluminum tubes (round, flat, and rectangular cross-sections) reveal that the material flow behavior under GLLF, with thickening primarily concentrated in the sidewalls and fillet regions—reaching up to twice the average circumferential compression. Finally, deployment of a GLLF production system enabled the manufacturing of e-bike tubular components with a significant improvement in efficiency, underscoring its potential for sustainable and high-efficiency production. Beyond practical benefits, this work establishes fundamental advances in pressure-volume coupling laws, critical pressure criteria, and deformation-flow mechanisms, offering a transferable framework for process design across diverse tubular geometries and materials.

1. Introduction

Lightweighting has emerged as a major trend in modern advanced manufacturing [1]. Material-level approaches using high-strength aluminum alloys [2], magnesium alloys [3], and novel layered composites [4] are now commonplace in aerospace, automotive, and related industries. Concurrently, structural solutions such as complex thin-walled metallic components [5] are increasingly used in high-end equipment. Fabrication methods for these intricate parts include laser additive manufacturing [6], hydroforming [7], spinning [8], electromagnetic forming [9], deep drawing [10], bending [11], and incremental sheet forming [12]. Hydroforming remains a cornerstone technique for producing complex, thin-walled integral tubular geometries.

Based on the prevailing deformation mechanism, tube hydroforming is classified into two types: high-pressure bulging [13], in which

circumferential tensile strains dominate and produce cross-sectional expansion; and low-pressure shaping [14], which is governed by circumferential compression and results in cross-sectional contraction. High-pressure bulging primarily includes conventional hydroforming at room temperature [15] and hot gas forming at elevated temperatures [16]. Among these, hot gas forming is particularly suitable for difficult-to-form materials such as titanium alloys [17]. However, high-pressure bulging typically requires large-scale equipment to provide sufficient die-closing force or high-energy heating furnaces to raise the workpiece to extremely high temperatures for forming [18].

Low-pressure tube shaping technology relies on the combined action of die movement and internal support pressure to achieve part formation under ambient temperature and low-pressure conditions. Compared to high-pressure bulging, this method offers significant advantages in terms of equipment tonnage, energy consumption, and process complexity. Based on the extent of circumferential compression, low-pressure tube forming can be classified into two categories: constant-

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Nomenclature			
P_0	Initial pressure	t	Wall thickness
V_0	Initial volume of tube	P	Pressure
V_{g0}	Initial volume of gas	V	Volume of tube
V_{m0}	Initial gas molar volume	V_g	Volume of gas
ξ	Volume compressibility ratio	V_m	Gas molar volume
V_l	Liquid volume	k	Gas proportion
ω	Acentric factor	n	Amount of substance
C	Solubility volume	T	Temperature
R	Gas constant	T_c	Critical temperature
L_1	Flattening displacement	T_r	Reduced Temperature
l	Length of straight segment	L_2	Die displacement
r	Radius of corner	P_c	Critical Pressure
		R_0	Initial radius of tube

perimeter forming processes [19] and liquid-filled upsetting processes involving significant circumferential compression. Nikhare et al. [20] analyzed the magnitude of the die-closing force required for low-pressure forming and established a corresponding theoretical calculation model. When the tube geometry changes during forming, the deformation is mainly characterized by in-plane bending, typically resulting in the formation of several plastic hinges [21] in regions of stress concentration. The material bends around these hinges, which can lead to structural warping. Nikhare et al. [22] further investigated the internal support pressure necessary for low-pressure forming of thin-walled tube and developed a critical support pressure model to predict the minimum internal pressure required to suppress cross-sectional warping under constant perimeter conditions. During the cross-sectional compression of thin-walled tube, defects such as buckling and wrinkling are prone to occur [23], making it necessary to apply a higher support pressure to effectively suppress wrinkling. To address buckling and wrinkling in thin sheets under compressive loads, researchers have mainly adopted three approaches: linear eigenvalue analysis [24], finite element simulation [25], and energy-based methods [26]. In the study of compression-induced instability during thin-walled component forming, the energy method is widely adopted due to its clear theoretical basis and computational efficiency. Fan et al. [27] developed a buckling model based on the energy method, incorporating interfacial interactions and the effect of facing thickness to predict the critical wrinkling stress in ultra-thin high-strength aluminum alloy shells. In another study, the compression deformation and buckling behavior of materials during fluid pressure forming were analyzed [28], and a critical internal pressure model was established using the energy method. However, the model requires input from numerical simulations, increasing its computational complexity. The material flow behavior during hydroforming of laminated tube blanks was investigated by Feng and Han [29], and theoretical models were proposed for predicting critical wrinkling stress and critical support pressure during compression. Similarly, Chu et al. [30] derived a theoretical model for the critical support pressure of sidewalls in round-to-rectangular cross-sectional forming using the energy method. The model was validated through finite element simulations and experiments, revealing that the critical support pressure increases with forming depth and strongly depends on the material strength coefficient and strain hardening exponent, but is independent of tube thickness.

In low-pressure forming processes, the coordination between die displacement and internal support pressure is a critical factor that directly determines the forming quality. Currently, the support media used in most low-pressure forming systems are incompressible liquids, which cause significant pressure fluctuations even with minimal volume changes. Existing pressure regulation strategies, such as constant pressure control [31] or stepwise loading [32], often involve complex mechanisms and fail to produce an ideal pressure evolution curve. Since

the internal medium is passively compressed and its volume change is irregular, more time is required to adjust the pressure when fluctuations occur. In contrast, gases offer excellent compressibility [33], making them promising alternatives for pressure regulation. Lin et al. proposed a gas internal-pressure-supported bending process [34], in which gas is injected into the tube cavity during bending to provide internal support and suppress wrinkling in thin-walled high-strength steel tube. Gas forming and press hardening are widely used in automotive tube shaping. Rauter et al. [35] showed that process variables, especially gas pressure, significantly affect tube geometry, mechanical properties, and formability. Bach et al. [36] investigated the combination of hot metal gas forming and conductive heating for producing high-strength, complex curved components. By controlling the pressure curve, rapid forming of the components is achieved, with pressure loading reaching up to 50 MPa. However, in forming components with complex shapes and high compression ratios, early-stage support pressure must be low to avoid bulging and an increase in cross-sectional perimeter, while higher pressure is required later to prevent wrinkling. The low gas-filling efficiency and difficulty in fine-tuning the pressure make it challenging to achieve rapid pressure adjustment, which can lead to potential wrinkling defects when using gas as the support medium.

As shown in Fig. 1, conventional low-pressure forming with incompressible media leads to sudden pressure spikes and fluctuations and requires complex or slow regulation strategies (e.g., step loading), while using a single, high-compression-ratio gas medium gives rise to instability-induced defects such as buckling and wrinkling, ultimately reducing forming reliability and efficiency.

To overcome these limitations, this study introduces a gas-liquid hybrid low-pressure forming (GLLF) process that reduces pressure fluctuation and enhances efficiency by combining the filling capability of liquid with the compressibility of gas through tunable ratios. The main contributions are: (i) establishing a compression model based on the Soave-Redlich-Kwong (SRK) real-gas equation, incorporating gas solubility effects to characterize the behavior of the hybrid medium; (ii) experimentally determining critical support pressure curves and demonstrating wrinkle suppression by tailoring gas-liquid parameters; and (iii) developing a GLLF forming platform, where round-to-square tube tests validated the model, enabled defect-free forming, and revealed material flow mechanisms through wall-thickness analysis. These advances provide both mechanistic insight and practical guidance, laying the foundation for wrinkle-free, high-efficiency forming of thin-walled tubular components

2. Principle of tube gas-liquid hybrid low-pressure forming

In low-pressure forming of thin-walled metallic tube, the initially circular cross-section is reshaped into a complex non-circular profile, resulting in a reduction of internal volume. By combining highly

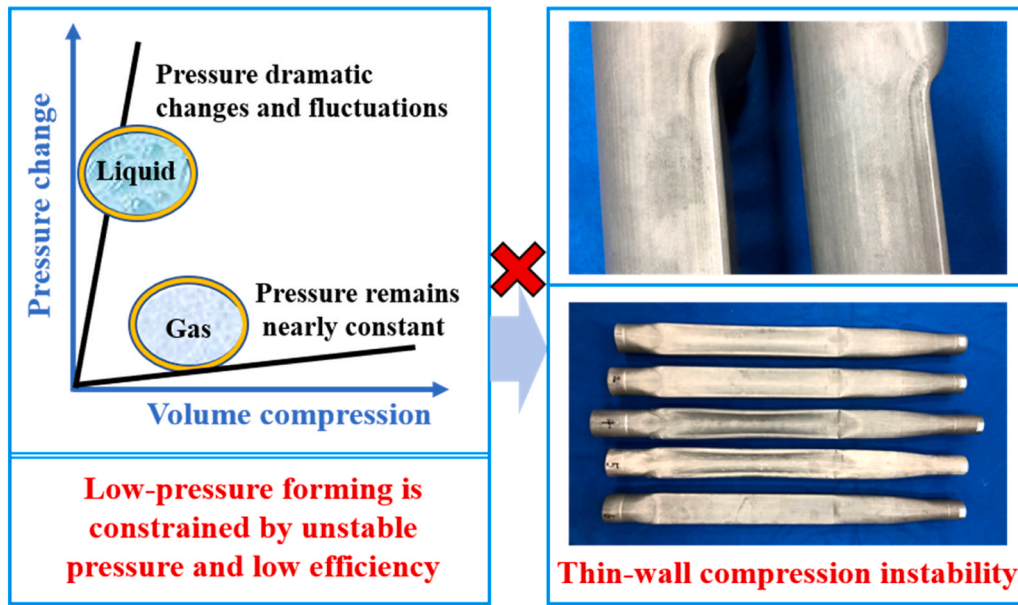


Fig. 1. The challenges and common issues faced by low-pressure forming technology.

compressible gas with nearly incompressible liquid, a gas-liquid hybrid medium with variable compressibility is formed, enabling smooth pressure evolution during volume reduction. Moreover, this self-pressurizing mechanism not only stabilizes the internal pressure but also mitigates compressive instability, thereby simultaneously overcoming pressure fluctuations and deformation-instability challenges in tube low-pressure forming.

The specific forming process, illustrated in Fig. 2a, comprises three distinct phases. First, liquid is injected into the tube cavity according to a predefined ratio; next, gas is introduced until the designated initial pressure P_0 is reached; finally, the die descends until full closure, rapidly completing the tube forming process. As shown in Fig. 2b, the pressure

evolution across distinct deformation stages exhibits the following characteristics: in the early stage, the deformation mainly involves geometric changes in the cross-section, requiring only a low level of support pressure. At this point, the gas-liquid mixture enters an initial compression phase, and the pressure increases slowly. As the forming depth increases and cross-sectional compression occurs, a higher internal support pressure is needed. The compressed gas then leads to a rapid pressure rise, which effectively suppresses instability-induced defects such as buckling and wrinkling. This demonstrates a strong correlation between the support pressure required by material compression and the pressure response of the gas-liquid hybrid medium, highlighting the clear advantages of using a compressible gas-liquid mixture as a support

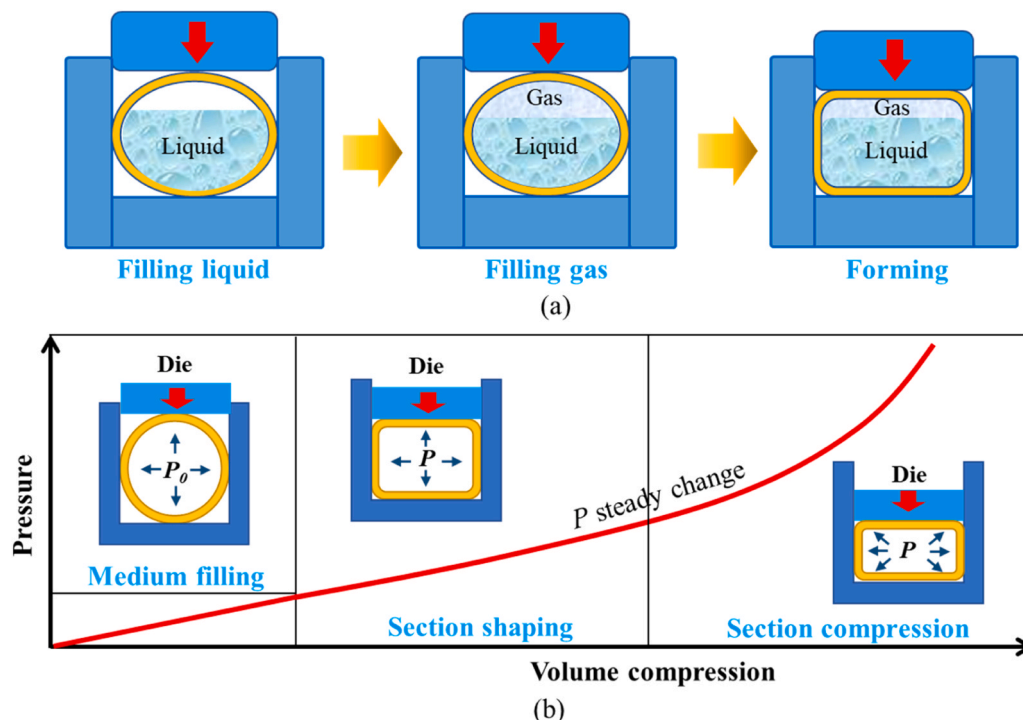


Fig. 2. Gas-liquid hybrid low-pressure forming process: (a) Process sequence, (b) Forming process with corresponding pressure evolution.

medium.

During the forming process, the internal pressure predominantly generated through gas compression, and its evolution is jointly determined by the initial gas volume fraction, initial pressure, and the volume compression ratio. By adjusting the initial pressure and the gas-liquid volume ratio, diverse pressure evolution curves can be generated in response to volume changes during forming, thereby meeting the requirements of different tube geometries. Fig. 3 presents the compression pressure-deformation curves for gas, liquid, and gas-liquid hybrid media under varying volume ratios and initial pressures, where, by assuming that the gas's volume compression equals the tube's volume reduction and applying the ideal gas law, the hybrid medium's compression behavior is derived. The results show that the gas-liquid volume ratio strongly influences the curve steepness: a lower gas fraction leads to a more rapid pressure rise with increasing compression ratio, while a higher gas fraction results in a more gradual trend approaching the liquid characteristics. Under the same gas-liquid ratio, increasing the initial pressure shifts the curves upward, thereby enhancing the load-bearing capacity of the medium. These findings demonstrate that by tuning the gas-liquid ratio and the initial pressure, diverse pressure evolution patterns can be obtained to meet the forming requirements of tubes with different geometries.

3. Theoretical model of gas-liquid hybrid medium compression

3.1. Fundamental theory of gas medium compression

The liquid used in our study is primarily water, which has a very small compressibility, approximately $4.5 \times 10^{-10}/\text{Pa}$. To illustrate, a pressure change of 1 Pa would reduce the volume of a given amount of water by only a factor of 4.5×10^{-10} . Assuming the liquid is completely incompressible, the pressure variation in the gas-liquid hybrid medium is primarily governed by the volume change of the gas phase, which can be described using a gas equation of state.

Gas behavior can be described by two main types of equations of state: the ideal gas law and real gas equations. The ideal gas law is simple in form and is commonly used for low-pressure conditions. In contrast, real gas equations are more complex and better suited for predicting gas properties under high-pressure conditions.

When the gas volume is compressed from V_0 to V , the volume compression ratio ξ is defined as:

$$\xi = \frac{V_0 - V}{V_0} \quad (1)$$

First, according to the ideal gas law, the pressure change after gas

volume compression can be calculated. Given an initial pressure P_0 , the corresponding pressure P at a volume compression ratio ξ is:

$$P = \frac{P_0 V_0}{V} = \frac{P_0}{1 - \xi} \quad (2)$$

Next, the effect of volume compression on pressure variation is calculated based on a real gas equation of state known as the Soave-Redlich-Kwong (SRK) real gas equation of state:

$$P = \frac{RT}{V_m - b} - \frac{a\alpha}{V_m(V_m + b)} \quad (3)$$

The parameters in Eq. (3) can be determined using the medium's critical pressure P_c , critical temperature T_c , gas constant R , and acentric factor ω , as shown in the following expressions:

$$\left\{ \begin{array}{l} a = \frac{0.42748R^2T_c^2}{P_c} \\ b = \frac{0.08664RT_c}{P_c} \\ \alpha = (1 + (0.48508 + 1.55171\omega - 0.15613\omega^2)(1 - T_r^{0.5}))^2 \\ T_r = \frac{T}{T_c} \end{array} \right. \quad (4)$$

Further, Eq. (3) can be rearranged into a cubic polynomial in terms of the molar volume of the gas, V_m :

$$PV_m^3 - RTV_m^2 - (Pb^2 + RTb - \alpha a)V_m - \alpha ab = 0 \quad (5)$$

By solving the above cubic equation using the initial state parameters, the initial molar volume of the gas, V_{m0} , can be obtained. When the volume compression ratio is ξ , the molar volume becomes V_m , and the relationship is given by:

$$\xi = \frac{V_{m0} - V_m}{V_{m0}} \quad (6)$$

According to the law of mass conservation, the molar volume V_m in Eq. (6) can be expressed in terms of the initial molar volume V_{m0} and the volume compression ratio ξ . Substituting this relationship into Eq. (3), the pressure P corresponding to a compression from initial pressure P_0 and initial molar volume V_{m0} at a compression ratio ξ can be calculated as:

$$P = \frac{RT}{(1 - \xi)V_{m0} - b} - \frac{a\alpha}{(1 - \xi)V_{m0}[(1 - \xi)V_{m0} + b]} \quad (7)$$

Thus, the relationship between gas pressure and volume compression

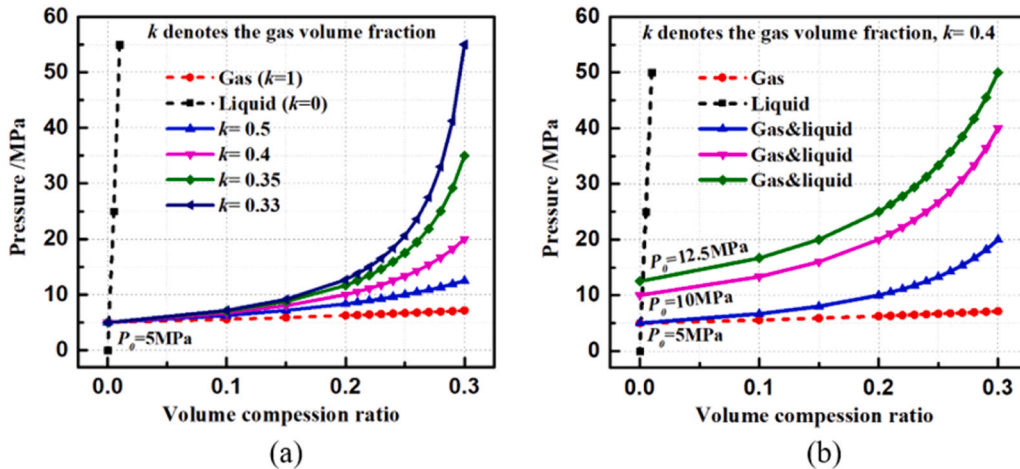


Fig. 3. Pressure-volume compression curves for gas, liquid, and hybrid medium: (a) Curves at varying gas volume fractions k with an initial pressure of $P_0 = 5 \text{ MPa}$, (b) Curves at varying initial pressures P_0 with a fixed gas volume fraction of $k = 0.4$.

ratio based on the SRK real gas equation of state is obtained.

By substituting the nitrogen physical property parameters into the pressure-volume compression ratio relationships derived from the ideal gas law and the SRK real gas equation, the compression pressure curves and corresponding pressure deviation curves of pure nitrogen were calculated under initial pressures of 1 MPa, 3 MPa, 5 MPa, and 8 MPa, as shown in Fig. 4. The results indicate that as the initial pressure increases, the discrepancy between the predictions of the two models gradually widens. At 1 MPa, a significant divergence between the curves occurs when the compression ratio exceeds 0.9. At 8 MPa, the two predictions begin to separate even at a compression ratio of 0.4. Moreover, as the compression ratio increases beyond a certain threshold, the pressure difference between the ideal gas and SRK models grows rapidly. For instance, at an initial pressure of 8 MPa and a compression ratio of 0.83, the pressure difference reaches 40 MPa, corresponding to a prediction error of 45 %.

Based on the above analysis, it can be concluded that the ideal gas equation is suitable for conditions involving low pressure and small compression ratios. However, as the pressure increases and the compression ratio becomes larger, the pressure deviation predicted by the ideal gas law grows significantly. Under such conditions, the real gas equation of state becomes necessary to accurately capture the pressure variation behavior during gas compression.

3.2. Compression model of gas-liquid hybrid medium

Based on the above analysis, this section develops a theoretical compression model for the gas-liquid hybrid medium, incorporating the gas solubility in the liquid phase and using the real gas equation of state [37]. A quantitative relationship is established between the current pressure and key parameters including the initial pressure P_0 , initial gas volume fraction k , and the volume compression ratio ξ of the mixed medium. The evolution of real gas equations includes the van der Waals equation, the Redlich-Kwong (RK) equation, and the Soave-Redlich-Kwong (SRK) equation [38]. In this study, the more accurate SRK equation of state is adopted to derive the compression model for the gas-liquid hybrid medium.

In a closed container, the amount of gas substance n_0 is equal to the ratio of the current gas volume V_g to the current molar volume V_m , expressed as:

$$n_0 = \frac{V_g}{V_m} \quad (8)$$

According to the law of conservation of mass, let the initial gas volume be V_{g0} , and the initial molar volume be V_{m0} . When the pressure changes from P_0 to P , the solubility of the gas in a liquid volume V_l is

considered. Let C denote the amount of gas dissolved per unit volume of liquid. Based on Henry's Law [39], and the current ratio of gas volume to molar volume can be expressed as:

$$\frac{V_g}{V_m} = \frac{V_{g0}}{V_{m0}} - (P - P_0)CV_l \quad (9)$$

Let the initial gas volume fraction be k , the volume compression ratio be ξ , and the initial total volume be V_0 , assuming that the liquid volume remains constant, we can derive the following relationships:

$$\begin{cases} \xi = \frac{V_{g0} - V_g}{V_0} \\ k = \frac{V_{g0}}{V_0} \end{cases} \quad (10)$$

By substituting the expressions for gas volume fraction k and volume compression ratio ξ from Eq. (10) into Eq. (9), we obtain:

$$\frac{(k - \xi)V_0}{V_m} = \frac{kV_0 - V_{m0}(P - P_0)C(1 - k)V_0}{V_{m0}} \quad (11)$$

Starting from the simplified form of Eq. (11), we get the expression for the current pressure P :

$$P = P_0 - \frac{(k - \xi)V_{m0} - kV_m}{V_m(1 - k)V_{m0}C} \quad (12)$$

Here, the initial molar volume V_{m0} can be obtained from Eq. (5) using the known initial state parameters. Therefore, once the current molar volume V_m is determined typically by solving the SRK real gas equation under the given compression ratio ξ the current pressure P can be calculated directly using the initial conditions and the derived expression for pressure as a function of ξ .

By substituting Eq. (12) the expression for pressure P as a function of V_m , ξ , and initial parameters into the SRK real gas equation of state (Eq. 5), a cubic polynomial in molar volume V_m can be obtained:

$$\begin{aligned} & \left[P_0 + \frac{k}{C(1 - k)V_{m0}} \right] V_m^3 - \left[\frac{k - \xi}{C(1 - k)} + RT \right] V_m^2 \\ & + \left[a\alpha - bRT - P_0b^2 - \frac{kb^2}{C(1 - k)V_{m0}} \right] V_m + \frac{k - \xi}{C(1 - k)}b^2 - ab\alpha = 0 \end{aligned} \quad (13)$$

By further solving the above cubic equation, the current molar volume V_m of the gas can be obtained. Substituting V_m back into Eq. (12) then yields the current pressure P of the gas-liquid hybrid medium under a given compression ratio ξ . When $\xi = 0$ (i.e., no compression has occurred), the system remains in its initial state, and we have $V_m = V_{m0}$, under this condition, Eq. (13) (the cubic polynomial form derived by combining the SRK equation and the compression model) degenerates to

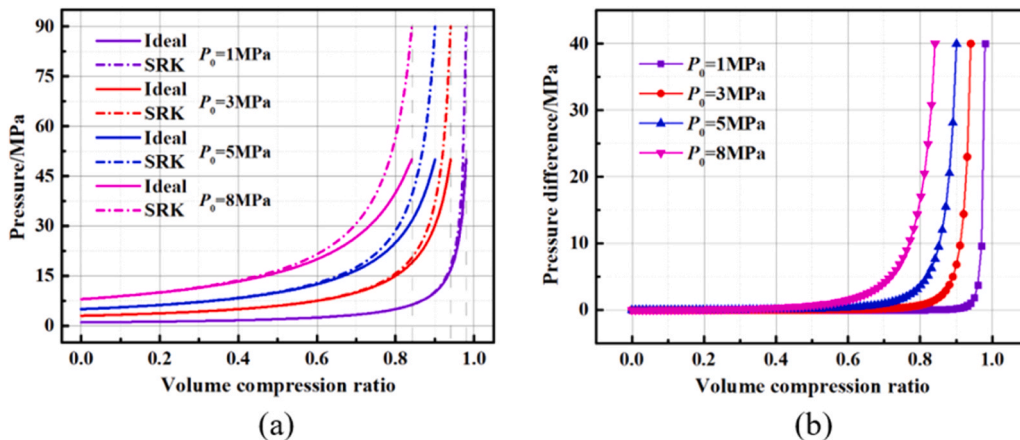


Fig. 4. Gas compression behavior at 298 K: (a) Pressure-compression curves using ideal-gas and SRK models at $P_0 = 1, 3, 5$, and 8 MPa, (b) Pressure deviation between models versus compression ratio.

Eq. (6), which is the original cubic equation in V_m describing the initial molar volume based on the initial pressure and SRK real gas behavior.

Thus, a compression model for the gas-liquid hybrid medium, incorporating gas solubility in the liquid phase and based on the SRK real gas equation of state, is established. This model provides a theoretical foundation for accurately calculating and analyzing the pressure evolution behavior of the internal medium during gas-liquid hybrid low-pressure forming of thin-walled tube.

3.3. Pressurization characteristics of gas-liquid hybrid medium

This study investigates how the initial pressure and gas volume fraction affect the pressure evolution of the gas-liquid hybrid medium under a specified compression ratio. At room temperature (298 K), the compression-induced pressurization behavior was analyzed based on the SRK-based gas-liquid compression model for a mixture of air (composed of nitrogen and oxygen in a 79:21 ratio) and water, with a maximum volume compression ratio of 0.45. By varying the gas volume fraction k and the initial pressure P_0 , the study systematically explores how these parameters influence pressure variation under a given compression condition. The model parameters used for the calculations, obtained from the National Institute of Standards and Technology (NIST) Chemistry WebBook, are listed in Table 1.

(1) Effect of gas volume fraction on the compression behavior of the gas-liquid hybrid medium

During tube forming, the volume reduction follows a similar trend across different conditions. Under the same overall volume compression ratio, a higher gas volume fraction results in smaller compression of the gas phase, leading to a relatively mild pressure increase. Conversely, a lower gas volume fraction causes the gas to undergo greater compression, resulting in a more significant pressure rise.

Using a mixture of air and water as an example, the influence of gas volume fraction on the compression pressure behavior of the gas-liquid hybrid medium was analyzed based on the theoretical model established in this study. As shown in Fig. 5a, under an initial pressure of 5 MPa, the compression pressure of the hybrid medium increases nonlinearly with the volume compression ratio ξ , exhibiting a slow-then-sharp rising trend.

The initial gas volume fraction k plays a decisive role in determining the magnitude of pressure increase. When $k = 0.90$, the pressure rises to only about 9 MPa at $\xi \approx 0.45$, which is approximately twice the initial pressure. Reducing k to 0.55 results in a final pressure exceeding 30 MPa (over six times the initial pressure). When k is further reduced to 0.5 and $\xi = 0.42$, the pressure reaches 40 MPa, representing more than eight times the initial value. These results highlight the strong sensitivity of pressure response to the gas-liquid volume ratio during compression.

Fig. 5b further quantifies the relationship between the final pressure and the gas volume fraction k under different volume compression ratios. As k increases, the final pressure exhibits a monotonic decrease. Additionally, the greater the ξ (i.e., the higher the curve), the more

sensitive the pressure is to changes in k . For instance, at $\xi = 0.4$, increasing k from 0.50 to 0.90 causes the final pressure to drop from approximately 28 MPa to 8 MPa, a difference of 20 MPa. In contrast, at $\xi = 0.1$, the same change in k results in a pressure difference of only about 1 MPa. This demonstrates that at higher compression levels, the gas-liquid ratio has a much stronger influence on the pressure response of the hybrid medium.

In summary, the gas-liquid hybrid medium shows a slow-then-sharp nonlinear pressure response during compression. As the volume compression ratio ξ increases, the pressure rise accelerates and is significantly affected by the initial gas volume fraction k . Lower k values yield higher final pressures, while higher k values flatten the pressure curve. At larger ξ , the system becomes more sensitive to k , amplifying its influence. Therefore, selecting an appropriate k based on the target ξ enables precise pressure control, enhancing forming stability and quality.

(2) Effect of initial pressure on the compression behavior of the gas-liquid hybrid medium

During the compression of a gas-liquid hybrid medium, the initial pressure not only determines the starting level of the pressure curve but can also significantly influence the rate of subsequent pressure increase. Using an air-water mixture as an example, this study employs the proposed gas-liquid hybrid compression model to systematically analyze the pressure evolution under various initial pressure conditions through theoretical calculations.

At a fixed initial gas volume fraction $k = 0.5$, the initial pressure P_0 was varied from 1 MPa to 8 MPa in 1 MPa increments. As shown in Fig. 6a, increasing P_0 significantly raises the final pressure under the same ξ . For example, when P_0 is 1 MPa and 3 MPa at $\xi = 0.45$, the final pressures reach approximately 10 MPa and 35 MPa, respectively, both achieving more than a tenfold increase. Furthermore, when P_0 is increased to 4, 6, and 8 MPa, the ξ required to reach a final pressure of 40 MPa is reduced to 0.438, 0.407, and 0.376, respectively. This indicates that higher initial pressure enables the same or even greater pressure levels to be achieved at lower compression ratios. Fig. 6b further illustrates that, under fixed ξ conditions, the final pressure increases approximately linearly with P_0 . For instance, at $\xi = 0.3$, increasing P_0 from 4 MPa to 8 MPa raises the final pressure from about 10 MPa to 20 MPa, effectively doubling it. Moreover, as ξ increases, the amplification effect of P_0 on the final pressure becomes more pronounced. These findings suggest that in tube forming design, careful coordination between the target compression ratio and the initial pressure is essential to achieve precise control over the pressure compression response of the gas-liquid hybrid medium.

In summary, with a constant gas volume fraction, the final pressure increases nearly linearly with the initial pressure. Higher initial pressure allows target pressures to be reached at lower compression ratios, and its amplification effect becomes more pronounced at larger ratios. Therefore, in tube forming design, the initial pressure should be carefully set based on the target compression ratio to ensure the desired pressure

Table 1
SRK equation of state parameters for nitrogen-water, oxygen-water and air-water mixtures.

Parameter	Unit	Nitrogen-water	Oxygen-water	Air- water
Critical temperature T_c	K	126.192	154.581	132.5
Critical pressure P_c	MPa	3.396	5.043	3.77
Acentric factor ω	–	0.0372	0.0222	0.011
Gas constant R	J/(mol·K)	8.31	8.31	8.31
System temperature T	K	298	298	298
Reduced temperature T_r	–	2.38	1.93	2.25
Alpha value α	–	0.498	0.637	0.5609
SRK parameter a	–	0.138	0.14	0.1376
SRK parameter b	–	2.675×10^{-5}	2.207×10^{-5}	2.53×10^{-5}
Correction coefficient C	mol/(L·MPa)	6.1×10^{-7}	1.3×10^{-6}	7.5×10^{-7}

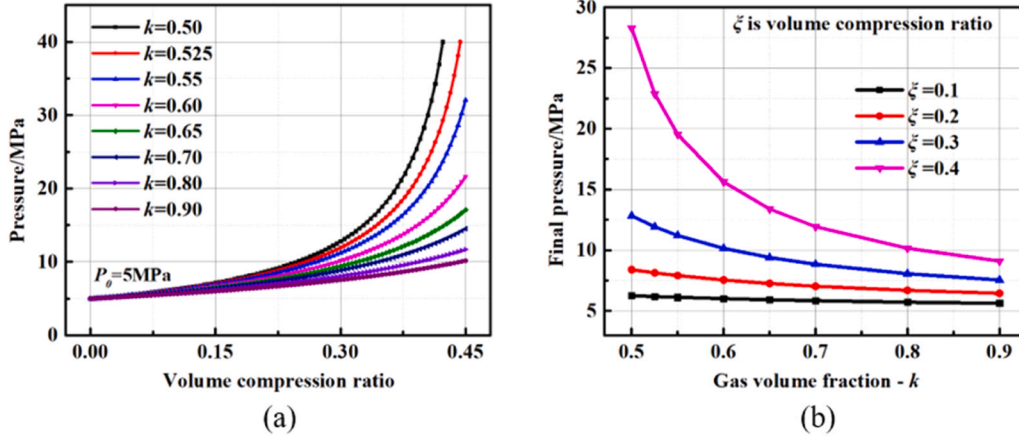


Fig. 5. Gas-liquid hybrid medium compression under $P_o = 5$ MPa: (a) Pressure-volume compression curves with different k ; (b) Final pressure vs. k for $\xi = 0.1, 0.2, 0.3$ and 0.4 .

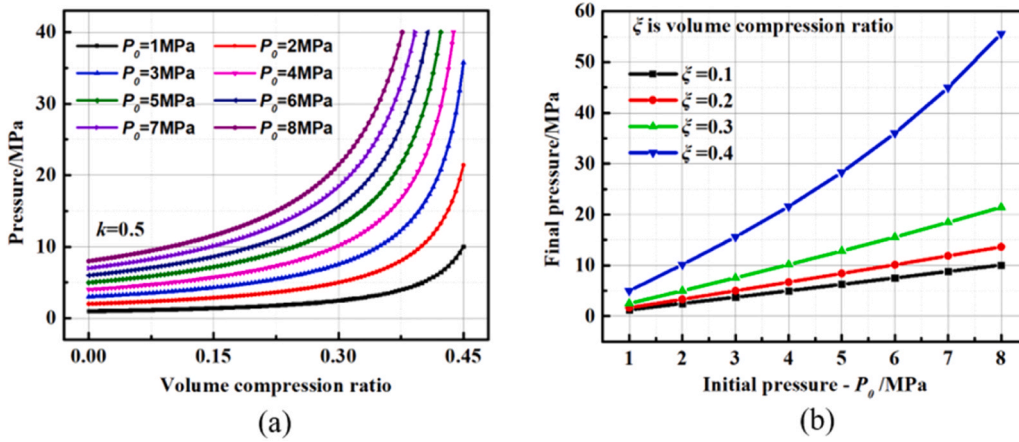


Fig. 6. Gas-liquid hybrid medium compression at $k = 0.5$: (a) Pressure-volume compression curves for $P_o = 1-8$ MPa, (b) Final pressure vs. P_o at $\xi = 0.1, 0.2, 0.3$ and 0.4 .

curve and stable forming.

- (3) Combined influence of gas volume fraction and initial pressure on pressure evolution

Both the gas volume fraction and the initial pressure are key parameters that can be independently adjusted to tailor the compression pressurization behavior of gas-liquid hybrid medium. By varying these two factors simultaneously, a broader and more flexible range of

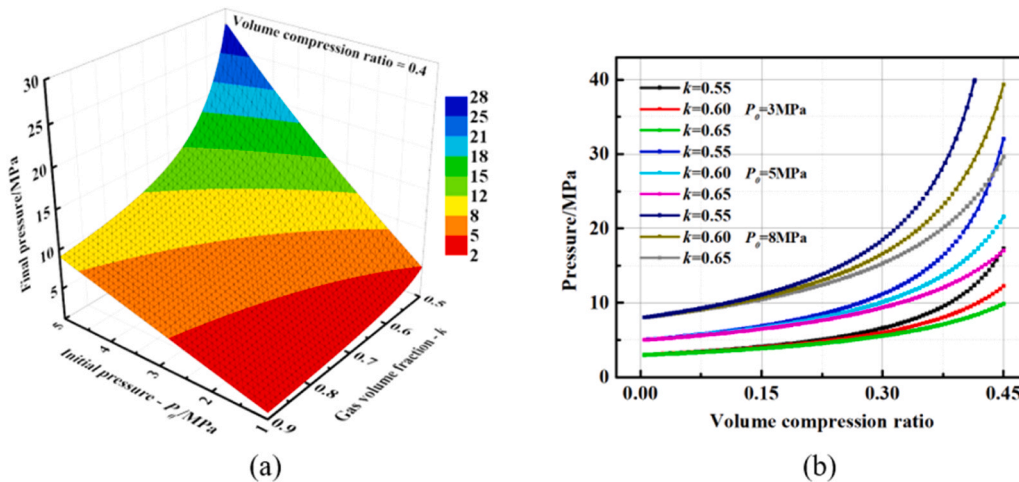


Fig. 7. Combined effects of gas fraction k and initial pressure P_o on gas compression behavior at 298 K: (a) 3D surface of pressure vs. ξ , P_o (1–5 MPa) and k (0.5–0.9), (b) Pressure- ξ curves at $P_o = 3, 5, 8$ MPa for $k = 0.55, 0.60$ and 0.65 .

pressure evolution curves can be achieved to meet different forming requirements. To further explore this effect, an air-water mixture is used as a representative example. Based on the theoretical gas-liquid compression model established in this study, a systematic analysis was conducted via theoretical calculations to investigate how the combined variation of gas volume fraction and initial pressure influences the pressure evolution during the compression process of the gas-liquid hybrid medium.

As shown in Fig. 7, the compression-induced pressurization behavior of the gas-liquid hybrid medium exhibits the following characteristics: The effect of initial pressure P_0 on the final pressure is approximately linear. For each 1 MPa increase in P_0 , the final pressure rises by 2–4 MPa, with higher sensitivity observed at lower gas volume fractions (e.g., $k = 0.50$). In contrast, the influence of gas volume fraction k on final pressure is both stronger and nonlinear. A decrease of 0.10 in k can result in a 4–8 MPa increase in final pressure. This gain becomes even more pronounced when either P_0 or the ξ is high. As ξ increases from 0.1 to 0.4, both P_0 and k exhibit amplified effects on pressure evolution, leading to larger pressure variations. Therefore, by coordinating the adjustment of initial pressure and gas volume fraction in accordance with the target compression ratio, it becomes possible to achieve precise control over the pressure curve's starting level, growth rate, and nonlinear turning points. This enables tailored pressure compression characteristics to meet the diverse forming requirements of different thin-walled tubular components.

In summary, the compression behavior of the gas-liquid hybrid medium is mainly influenced by initial pressure and gas volume fraction. Higher initial pressure raises the entire pressure curve, while lower gas fraction enhances compressibility, accelerating pressure increase. At larger compression ratios, both factors have amplified effects on pressure gain. Coordinated adjustment of these parameters enables precise control over the pressure curve, allowing customized pressure profiles to meet various thin-walled tube forming requirements.

4. Experiment

4.1. Experimental setup

As shown in Fig. 8, the experimental setup developed in this study for gas-liquid hybrid low-pressure forming of thin-walled metallic tube

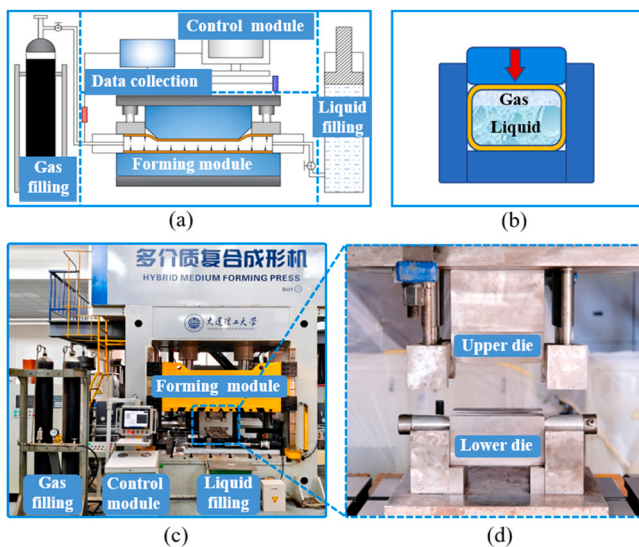


Fig. 8. Experimental setup for GLLF forming: (a) Schematic of the hybrid charging system (liquid and gas units), (b) Annular cross-section compression schematic, (c) Hybrid system hardware (charging units and control), (d) Forming die assembly on 200-ton servo press.

consists of four main components: the liquid charging unit, gas charging unit, control unit, and forming unit. The liquid charging unit comprises a servo motor and a liquid injection piston cylinder, which is connected to the tube cavity via a one-way valve to enable precise control of the injected liquid volume. The gas charging unit includes a high-pressure gas cylinder, electromagnetic valve, gas lines, and a pressure sensor, allowing for accurate regulation of the gas pressure. The forming unit is composed of a 200-ton servo press and a round-to-rectangular forming die, equipped with a laser displacement sensor to precisely record the die movement. The forming die includes end blocks and a mold cavity designed to perform round-to-rectangular deformation. All components are integrated through the centralized control unit, enabling coordinated operation and execution of the gas-liquid hybrid forming experiments for thin-walled tube.

4.2. Experimental forming procedure

This study uses 6061-O aluminum alloy thin-walled circular tube as the test material. The tube has an outer diameter of 38 mm, a wall thickness of 1.2 mm, and a length of 365 mm. To achieve a longer sidewall length in the final rectangular cross-section suitable for investigating compression-induced instability and material flow behavior in gas-liquid hybrid low-pressure forming, the tube is not formed directly from a circular to a rectangular shape. Instead, the tube is initially flattened into a racetrack-shaped (oval) cross-section without internal support. It is then further compressed along the direction perpendicular to the flattening axis, using a gas-liquid support medium, to form the final rectangular cross-section. As shown in Fig. 9, this two-step forming process (flattening followed by rectangular forming) enables the side-walls to undergo more significant compressive deformation, which better reflects the mechanical behavior of the material under low-pressure forming conditions. The geometric dimensions of the tube blank and the key parameters of the forming dies are summarized in Table 2.

5. Results and discussion

5.1. Compression behavior of the gas-liquid medium in low-pressure tube forming

An experimental setup was established in this study to investigate the round-to-flattened-to-rectangular forming of thin-walled 6061 aluminum alloy tube under gas-liquid hybrid medium support. A controlled mixture of air and water at varying volume ratios was injected into the tube cavity. By adjusting both the initial internal pressure and the gas volume fraction, different pressure evolution curves were generated to enable the forming of tube under various compression conditions. The effective internal volume of the tube was measured before and after forming to determine the volume change of the gas-liquid medium. These experimental volume changes, along with the

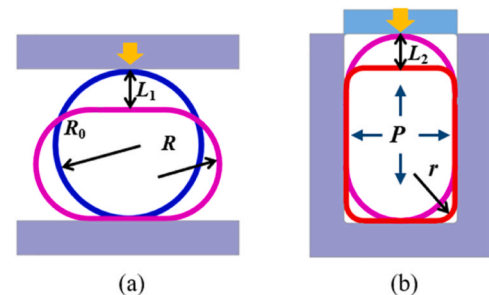


Fig. 9. Two-step tube forming: (a) Flattening to racetrack-shaped section without inner support, (b) Perpendicular compression to rectangular section with gas-liquid support medium.

Table 2

Geometric dimensions of tube blank and key die parameters.

Parameter	Symbol	Value	Unit
Tube wall thickness	t	1.2	mm
Outer radius of circular tube	R_0	19	mm
Flattening displacement	L_f	10	mm
Die corner radius	r	1.5	mm
Die width	–	28	mm

corresponding process parameters, were then substituted into the gas-liquid hybrid compression model developed in this study, which is based on the SRK real gas equation of state. The resulting theoretical pressure predictions were compared against the experimental data to evaluate the model's reliability and accuracy. The air used in the experiments consisted of 79 % nitrogen and 21 % oxygen, with the relevant thermophysical parameters listed in Table 1.

As shown in Fig. 10, a series of round-to-rectangular forming experiments were conducted on thin-walled tube under varying compression amounts. The internal volume of the tube specimens corresponding to different forming stages was measured using the water-filling method, and the results are summarized in Table 3. Using these measured volumes, the volume compression ratios were calculated and substituted into the proposed gas-liquid hybrid compression model to obtain the theoretical internal pressure under different gas-liquid ratios and initial pressures. During forming, the internal pressure was recorded using pressure sensors, and the die displacement was monitored with a laser displacement sensor, allowing the construction of measured pressure–displacement curves.

As shown in Fig. 11, which compares the measured displacement–pressure curves with theoretical predictions, the results reveal a strong agreement in both trend and magnitude. Using the experimental data as a benchmark, the theoretical internal pressure values were calculated for different die displacements $L_2 = 14$ –18 mm, corresponding to various volume compression ratios. These theoretical values were then compared with the measured pressures to assess model accuracy. Fig. 11a presents the die displacement–pressure curves under multiple test conditions. Experimental data are marked by red rectangular, fitted curves by purple lines, and theoretical predictions calculated using the SRK real gas equation of state are shown as blue stars. The results clearly show that the trend and magnitude of the theoretical predictions align well with the experimental data across all test conditions. This indicates that the model reliably captures the pressure evolution of the gas-liquid hybrid medium during compression. Notably, the model maintains stable prediction performance under varying gas volume fractions k and initial pressures P_0 . Fig. 11b further quantifies the percentage error

Table 3

Forming parameters and volume changes of tube specimens.

Die displacement L_2 (mm)	0	14	15	16	17	18
Medium volume V (ml)	305	265.1	259.2	253.0	246.9	240.6
Gas volume fraction k	–	0.40	0.35	0.30	0.25	0.25
Initial pressure P_0 (MPa)	–	1.8	2.0	2.0	2.0	2.5
Final pressure (MPa)	–	2.5	3.3	4.6	7.8	15.1

between theoretical and experimental values across different compression displacements. The results show that at $L_2 = 15$ mm, the model achieves its lowest average error of just 0.44 %, while the highest average error of 2.17 % occurs at $L_2 = 18$ mm. Overall, the low mean deviation observed under all forming conditions highlights the model's strong predictive accuracy.

In conclusion, the gas-liquid hybrid compression model developed in this study based on the SRK real gas equation of state exhibits excellent agreement with experimental data under various die displacements, initial pressures, and gas volume ratios. This verifies the model's reliability and applicability in the design and control of low-pressure gas-liquid hybrid forming processes.

5.2. Critical support pressure for compression instability in gas-liquid forming

In this section, an experimental method was developed to measure the critical support pressure associated with compression-induced instability in the straight-wall segment of thin-walled tubes. Round-to-flattened-to-rectangular forming experiments were conducted using 6061-O aluminum alloy tubes. The relationship between die displacement and critical support pressure was experimentally measured under different cross-sectional compression conditions of the tube blank.

First, a series of round-to-flattened-to-rectangular forming experiments were conducted on thin-walled tubes under varying die displacements. The gas volume fraction k was fixed at 0.5, and the internal pressure evolution during compression was controlled by adjusting the initial gas pressure. By setting different die displacements, specimens with varying circumferential compression ratios were obtained. These formed tubes exhibited instability behaviors (such as wrinkling) under different compression levels, providing the basis for analyzing the relationship between die displacement and critical support pressure.

When no instability occurs, the circumferential perimeter of the tube cross-section decreases continuously with increasing die displacement. However, once buckling or wrinkling initiates, the die displacement continues to increase while the circumferential perimeter remains constant. To determine the onset of instability, GOM Scan 1 laser scanning equipment was used to obtain the full surface contour of the wrinkled tube specimens, with a scanning accuracy of up to 0.005 mm. The circumferential cross-sectional perimeter was then extracted from the scanned geometry. Based on the initial tube dimensions and die geometry parameters (as listed in Table 2), the theoretical perimeter corresponding to each die displacement was calculated. The difference between the theoretical and measured perimeter, divided by two, represents the displacement gap between the theoretical perimeter reduction and the actual onset of wrinkling. This allows the identification of the critical die displacement at which instability (wrinkling) begins. Then, using the experimentally recorded displacement–pressure curve, the internal pressure corresponding to the onset of wrinkling can be identified, which is defined as the critical support pressure at that specific die displacement.

As shown in Fig. 12a, the relationship between die displacement and internal pressure under different initial pressures during the gas-liquid hybrid forming process. The results show that as die displacement increases, the internal pressure rises continuously until reaching a critical

**Fig. 10.** Experimental results at different die displacements.

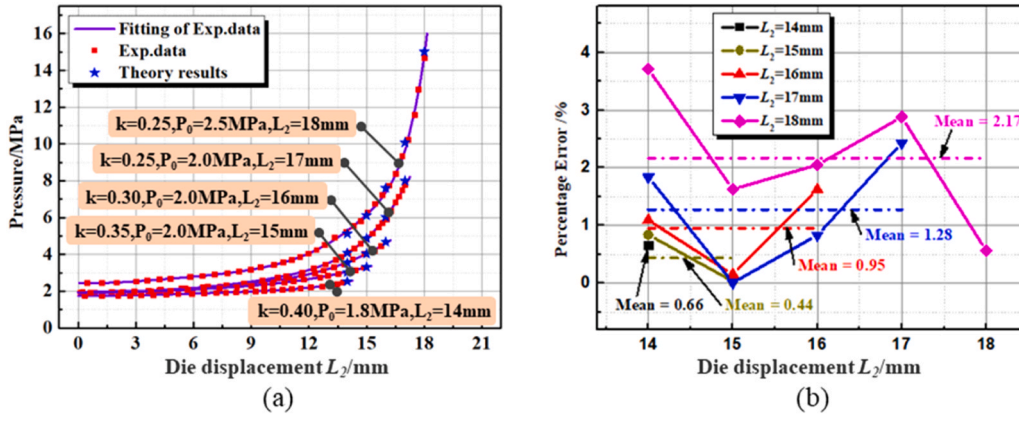


Fig. 11. Theoretical vs. experimental pressure–displacement results for $L_z = 14\text{--}18\text{ mm}$ under $k = 0.25\text{--}0.40$ and $P_0 = 1.8\text{--}2.5\text{ MPa}$: (a) Shows the experimental pressure curves, fitted results, and theoretical predictions, (b) Displays the corresponding percentage errors between theoretical and experimental values.

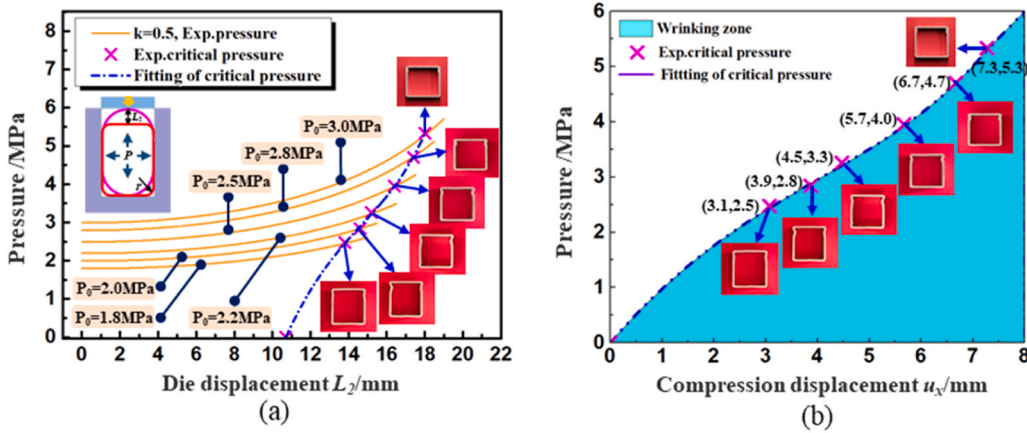


Fig. 12. Critical support pressure in gas-liquid low-pressure forming: (a) Experimental critical support pressure, (b) Tube compression forming instability zone delineation.

point, beyond which instability occurs. With higher initial pressures, the pressure–displacement curves shift upward, and the corresponding critical pressure also increases, indicating that larger initial pressures enhance the load-bearing capacity of the tube. The experimentally measured critical pressures agree well with the fitted results, confirming the reliability of the proposed model. The cross-sectional profiles embedded along the curves further demonstrate the deformation evolution from circular to square shapes, highlighting the progressive thinning and instability in the straight-wall regions as displacement increases. These results reveal that both die displacement and initial pressure play key roles in governing the onset of instability, thereby providing a clear guideline for selecting suitable forming parameters.

In the round-to-flat-to-square deformation of thin-walled tubes, the initial phase involves bending with a constant perimeter. The primary deformation occurs in the corners, where the radius decreases as displacement increases. Once the corner radius matches the mold's corner radius, further bending is not possible, and the tube transitions to cross-sectional compression. Assuming the perimeter remains constant and the cross-section transforms from circular to a rounded rectangle, it was calculated that when $L_z = 10.73\text{ mm}$, the compression displacement u_x begins to increase from zero, i.e., $u_x = L_z - 10.73$. Fig. 12b illustrates the relationship between compression displacement and internal pressure, highlighting the experimentally determined critical pressures and the corresponding wrinkle suppression window. The results indicate that as compression displacement increases, the critical supporting pressure rises nonlinearly, implying that greater deformation requires higher internal support to prevent instability. The shaded wrinkling

zone clearly demonstrates that when the internal pressure is lower than the critical supporting pressure, wrinkling defects inevitably occur, whereas maintaining the pressure above this threshold enables effective suppression of such defects. The experimental critical pressures are in good agreement with the fitted curve, confirming the validity of the proposed criterion. Furthermore, the embedded cross-sectional profiles illustrate the progressive development of wrinkling with increasing displacement, thereby providing direct experimental evidence for the identified instability mechanism.

In conclusion, the experiments establish a reliable method to determine the critical support pressure for thin-walled tubes. Results show that both die displacement and initial pressure significantly influence the onset of instability, with the critical support pressure increasing nonlinearly with compression ratio. These findings provide practical guidelines for optimizing parameters in gas-liquid hybrid low-pressure forming to achieve stable, wrinkle-free deformation.

5.3. Suppressing thin-wall instability by adjusting gas-liquid mixing parameters

By adjusting the gas-liquid ratio and the initial internal pressure of the gas-liquid hybrid medium, the internal support pressure within the tube can be effectively controlled to suppress the wrinkling instability commonly observed during material compression. Building upon the previously determined critical support pressure through experiments (Fig. 12), this section further explores how tailoring process parameters can regulate the internal pressure to prevent instability. Under a

constant die displacement, the gas volume fraction was reduced to 0.35, and the corresponding die displacement-pressure curve was recorded. As shown in Fig. 13a, a group of defect-free, wrinkle-free round-to-rectangular formed tube specimens were successfully obtained. The pressure curve clearly illustrates that when the internal pressure remains below the critical support pressure, wrinkling initiates; however, when the internal pressure exceeds the critical threshold, the tube forms successfully without instability. These results confirm that by modulating the gas-liquid hybrid forming parameters, it is possible to shape the internal pressure profile to match the evolving deformation requirements. This enables precise suppression of wrinkling defects, demonstrating excellent pressure control flexibility and effectiveness of the gas-liquid hybrid forming strategy. Fig. 13b illustrates the forming window and stability characteristics of the gas-liquid hybrid low-pressure forming process. It further divides the window into wrinkling, critical, and safe zones, clearly defining the pressure-displacement range required to ensure stable, wrinkle-free forming.

In gas-liquid hybrid low-pressure forming, maintaining internal pressure above the critical threshold is essential to prevent wrinkling. As shown in Fig. 13b, once the pressure drops below this limit, wrinkling occurs and becomes irreversible, even with subsequent pressure increases. Two key parameters influence pressure behavior: initial pressure and gas-liquid ratio. Increasing the initial pressure raises the overall pressure level but has limited effect on the pressure rise rate, which may still fall short in later stages (e.g., $k = 0.5$). In contrast, reducing the gas volume fraction (e.g., $k = 0.35$) significantly steepens the pressure curve, enabling it to quickly surpass the critical pressure and effectively suppress wrinkling. However, too little gas may cause over-forming or local thickening. Thus, optimal forming requires balancing the gas-liquid ratio and initial pressure. A well-designed pressure path, aligned with material deformation, ensures wrinkle-free, stable, and uniform tube forming.

In conclusion, the experimental results demonstrate that maintaining the internal pressure above the critical support threshold is the key to suppressing wrinkling in gas-liquid hybrid low-pressure forming. By carefully adjusting the initial pressure and gas-liquid ratio, the pressure evolution can be tailored to match the deformation process, thereby achieving stable, wrinkle-free, and uniform tube forming. This highlights the flexibility and effectiveness of the gas-liquid hybrid strategy in precisely regulating internal support pressure and provides clear guidelines for optimizing forming parameters in practical applications.

5.4. Wall thickness distribution and material flow in low-pressure tube forming

In this study, 6061-O aluminum alloy thin-walled circular tubes were

used as the research object. By controlling the die displacement during low-pressure forming, experimental results were obtained under various circumferential compression ratios. Analysis of wall thickness variations in the formed specimens provides valuable insight into the material flow behavior during the forming process. As shown in Fig. 14, the rectangular cross-section of the formed tube was divided into distinct regions. A total of 24 measurement points were selected: 5 points evenly distributed along each of the four flat sides, and 4 points positioned in the corner regions. Wall thickness measurements at these locations were used to characterize the distribution of wall thickness throughout the cross-section during the round-to-flattened-to-rectangular forming process. This data enables a more detailed understanding of material flow patterns, particularly how material redistributes under the combined effects of low internal pressure and die displacement during compression.

As shown in Fig. 15, the wall thickness distribution of thin-walled tubes during gas-liquid hybrid low-pressure forming exhibits distinct trends with increasing circumferential compression. The top and bottom regions remain relatively stable, with minimal thinning or thickening ($<5\%$), indicating limited deformation and little contribution to material redistribution. In contrast, the sidewalls and especially the corners display significant progressive thickening. At lower compression ratios ($\approx 5.5\%$, $L_2 = 14$ mm), sidewall thickening reaches about 11 % and corner regions 21–29 %, while at higher compression ratios ($\approx 12.2\%$, $L_2 = 18$ mm), sidewall thickening increases to about 24.5 % and corner thickening reaches 35–41 %. This reflects a consistent material flow path: material migrates from the relatively stable top and bottom walls toward the sidewalls and corners, where compressive strain is concentrated. The corners, due to geometric confinement and multi-axial loading, act as primary accumulation zones, resulting in the most severe thickening. Overall, the results demonstrate that the GLLF process

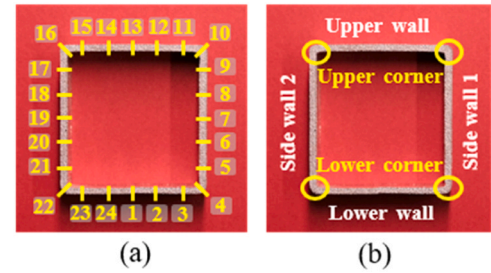


Fig. 14. Schematic diagram of rectangular cross-section partitioning: (a) Wall thickness measurement point locations, (b) Regional division of the rectangular cross-section.

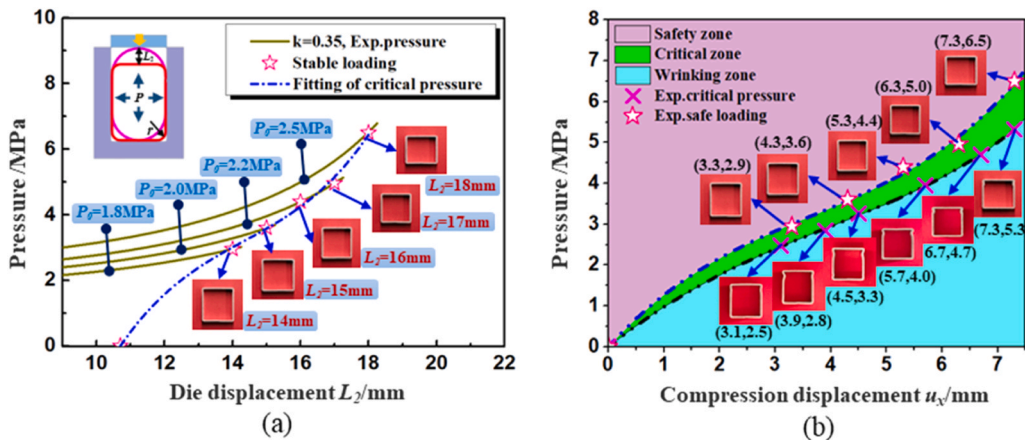


Fig. 13. Regulation of internal support pressure in tube compression forming: (a) Effect of gas-liquid ratio $k = 0.35$ on suppressing wrinkling, (b) Compression forming instability-zone delineation.

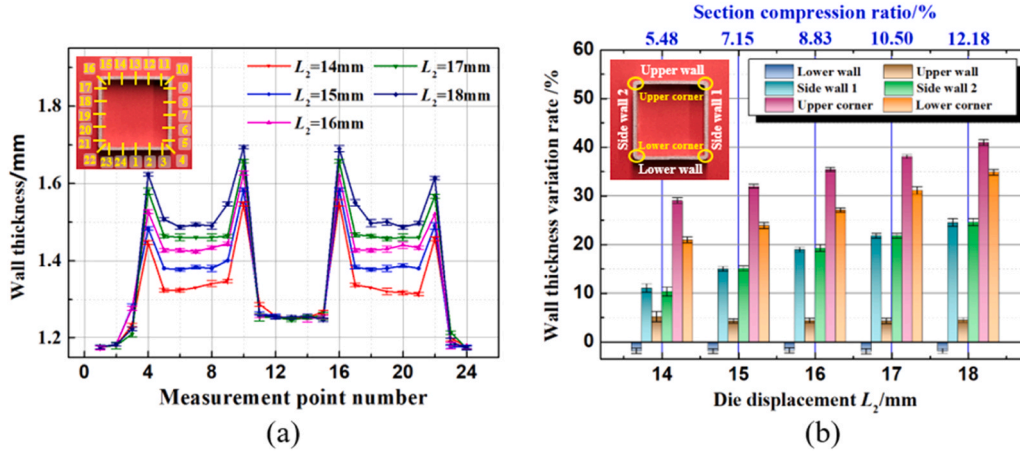


Fig. 15. Wall thickness evolution in GLLF forming ($L_2 = 14\text{--}18\text{ mm}$, $k = 0.35$, $P_0 = 2.0\text{ MPa}$): (a) Thickness distribution across regions at $L_2 = 14\text{--}18\text{ mm}$, (b) Percentage thickness change vs. L_2 (sectional compression ratio) for top, bottom, sidewall, and corner regions.

provides effective internal support for controlled deformation, while wall thickening correlates linearly with die displacement in key regions. These insights are crucial for die design, corner radius optimization, and pressure-path planning to ensure uniform thickness and high-quality forming of thin-walled components.

When the cross-sectional perimeter remains constant, geometric analysis shows that once the die displacement L_2 reaches 10.73 mm, the tube corner region conforms to the die corner radius of 1.5 mm. Beyond this point, deformation enters the cross-sectional compression stage. As the die displacement continues to increase, the top and bottom wall thicknesses remain unchanged, while only the sidewalls undergo further circumferential compression.

As shown in Fig. 16, the relationship between total circumferential compression ratio and sidewall compression ratio reveals that the sidewall compression ratio is approximately twice the overall circumferential compression ratio. This indicates that deformation becomes increasingly localized in the sidewall region during the later stages of forming. From this analysis, it can be concluded that during the early stage of deformation, as the shape evolves, plastic hinges are likely to form particularly at the corner regions, which serve as transition zones. These hinges significantly influence material redistribution, leading to the observed differences in wall thickness between the top and bottom surfaces. Once the corners are fully formed, they act as boundaries that restrict material flow, making it difficult for the material to cross from one region (e.g., sidewall) to another (e.g., top or bottom). As a result, subsequent material flow and thickening are concentrated primarily in the sidewalls and corners, with the corners exhibiting the maximum wall thickening due to localized compression and geometric confinement.

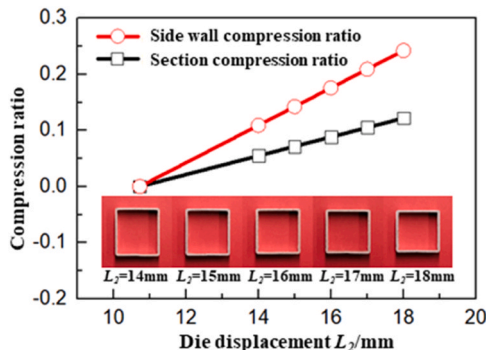


Fig. 16. Circumferential cross-section and sidewall compression ratio.

6. Application

To address the limitations of conventional high-pressure tube hydroforming—which demands extreme internal fluid pressures and imposes stringent requirements on equipment and die strength—this study introduces an ambient-temperature, low-pressure forming paradigm. Central to this approach is a gas-liquid hybrid medium with tunable compressibility: liquid and compressible gas are co-injected into the tube cavity in adjustable proportions, enabling a smooth, programmable pressure-volume response. This design not only lowers forming pressures compared to traditional methods but also overcomes core challenges of low-pressure forming, such as pressure regulation difficulty and low process efficiency.

Guided by theoretical modeling and validated experimentally, we developed a custom gas-liquid hybrid low-pressure forming system (Fig. 17a), comprising liquid and gas charging units, a forming chamber, and a high-precision control module. By tuning the initial pressure and gas volume fraction, the system precisely matches internal support pressure to the deformation sequence, effectively suppressing compression-induced instabilities (e.g., wrinkling) and enhancing both form quality and process robustness.

This equipment has already achieved substantial success in manufacturing bicycle and premium e-bike frame components. As shown in Fig. 17b-c, it has been used to produce tubular part variants with stable, efficient operation—delivering a great boost in production efficiency over conventional methods. Moreover, the technology offers a viable route to forming ultra-large, thin-walled parts that are otherwise infeasible with high-pressure processes. By combining low-pressure safety with high throughput, gas-liquid hybrid forming paves the way for low-energy, green production in line with the goals of sustainable, intelligent manufacturing.

7. Conclusions

To address the challenges of pressure fluctuation and low efficiency in conventional low-pressure forming, a novel gas-liquid hybrid forming method was developed and its theoretical and experimental performance for thin-walled tube forming was validated. The main conclusions are as follows:

- (1) A comprehensive compression model for gas-liquid mixtures was developed based on the Soave-Redlich-Kwong (SRK) real gas equation of state, which incorporates the dissolution behavior of gas in liquid. The model analyzed the impact of key factors, including initial gas volume fraction, initial pressure, and volume compression ratio, on pressure evolution, with diverse variations

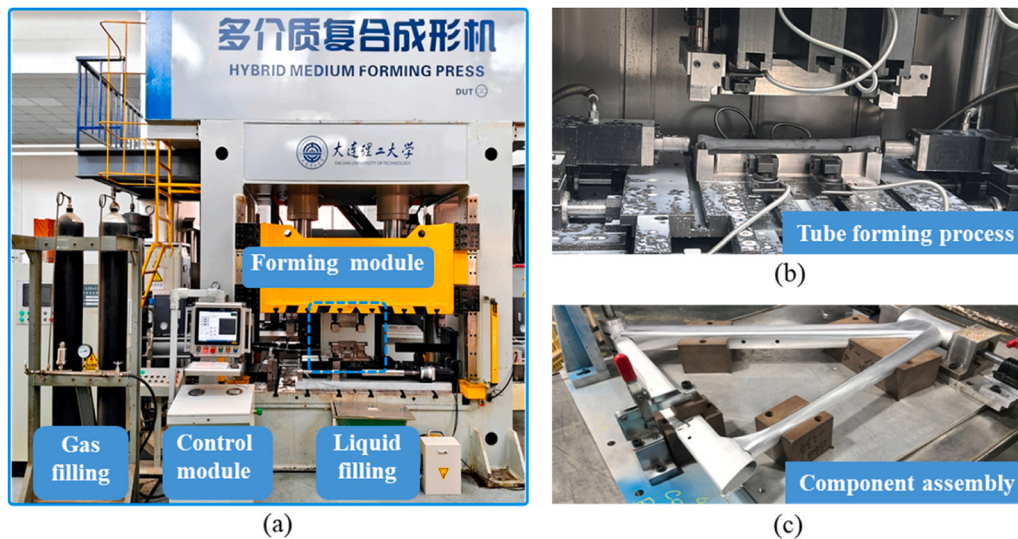


Fig. 17. GLLF system and industrial deployment: (a) Custom-designed forming equipment, (b) Representative e-bike tube forming process of GLLF, (c) Assembled e-bike frame module.

achievable by adjusting the initial pressure and gas fraction. Experimental validation demonstrated the model's reliability, with a maximum average prediction error of only 2.17 %, confirming its high accuracy and practical applicability in gas-liquid hybrid forming processes.

- (2) A new strategy for effective wrinkle suppression was proposed, focusing on dynamically adjusting the gas-liquid volume fraction and initial gas pressure during the forming process. Experimental results showed that at an initial pressure of 1.8–2.5 MPa, maintaining a gas fraction of 0.5 led to wrinkling under varying compression. However, reducing the gas fraction to 0.3 effectively suppressed the wrinkles. This dynamic adjustment maintained internal support pressure above the critical threshold, preventing wrinkling without overly conservative pressure settings, offering a more practical and efficient solution for wrinkle suppression in low-pressure forming.
- (3) The wall thickness analysis of the tube formed at different displacements showed that thickening occurred mainly in the sidewalls and corners. At a cross-sectional compression ratio of 5.5 %, the average thickening was 11 % in the sidewall and 25 % in the corner; at 12.2 %, thickening increased to 24.5 % and 38 %, respectively. The thickness variation was primarily influenced by initial material distribution and cross-sectional compression ratio. Once the corners were fully formed, material flow became restricted, resulting in localized thickening. The sidewall compression ratio was about twice the overall circumferential compression ratio.
- (4) A dedicated gas-liquid hybrid low-pressure forming system was developed, integrating a 200-ton servo press, liquid and gas filling units, a high-precision control module, and forming molds. This system enabled the efficient production of e-bike tubular components. By utilizing the gas-liquid hybrid medium, the system effectively suppressed wrinkling during compression. Compared to traditional methods, it significantly improved production efficiency, reduced energy consumption, and demonstrated the great potential of gas-liquid hybrid low-pressure forming technology for sustainable and high-efficiency manufacturing.

While this study systematically investigated the compression behavior of the gas-liquid medium and the instability mechanisms during low-pressure forming, and demonstrated clear advantages and

promising application potential, further research is still needed in the following areas: (1) A more comprehensive analysis of the geometrical evolution of the cross-section during forming, especially the behavior and forming limits of the corner region, is needed to reveal the forming capability of the low-pressure forming process. (2) Volume-pressure coupling during compression increases uncertainty and complicates process optimization. Therefore, there is an urgent need for finite element methods and refined design methods to optimize and establish process parameters for complex thin-walled tubes in gas-liquid hybrid low-pressure forming. (3) To expand manufacturing capabilities for a broader range of components especially large and extra-large parts and to promote the continued transformation and upgrading of manufacturing technologies.

CRediT authorship contribution statement

Hang Yuan: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation. **Gaoning Tian:** Writing – review & editing, Investigation, Formal analysis. **Yanli Lin:** Writing – review & editing, Writing – original draft. **Yi Xu:** Investigation, Formal analysis. **Zhijie Hu:** Validation, Data curation. **Kelin Chen:** Writing – review & editing. **Zhubin He:** Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jmatprotec.2025.119120](https://doi.org/10.1016/j.jmatprotec.2025.119120).

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